

# Integer part, fractional part and piecewise-defined functions

# Outline

Integer part and fractional part functions

Piecewise-defined functions

# Integer part and fractional part functions



# Introduction: Integer part and fractional part functions

- ▶ Two functions that are typically encountered when writing *algorithms* are the integer part (floor) function and the fractional part (or decimal part) function
- ▶ In this section we define them, together with the **upper integer part (ceiling)** function
- ▶ We draw their graphs: they are “step” functions, piecewise constant, and the fractional part is periodic

# Integer part and upper integer part

## Definition 1: of the integer part function

*The integer part (floor) function is:*

$$f : \mathbb{R} \rightarrow \mathbb{Z}, x \mapsto [x] \quad (\text{or } x \mapsto \lfloor x \rfloor)$$

## Definition 2: of the upper integer part function

*The upper integer part (ceiling) function is:*

$$f : \mathbb{R} \rightarrow \mathbb{Z}, x \mapsto \lceil x \rceil$$

## Definition 3: of the fractional part function

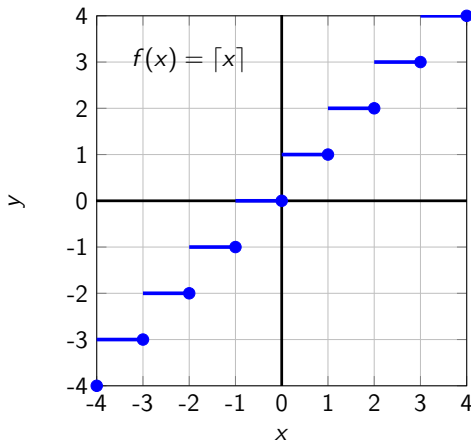
*The fractional part function is:*

$$f : \mathbb{R} \rightarrow [0, 1), x \mapsto (x)$$

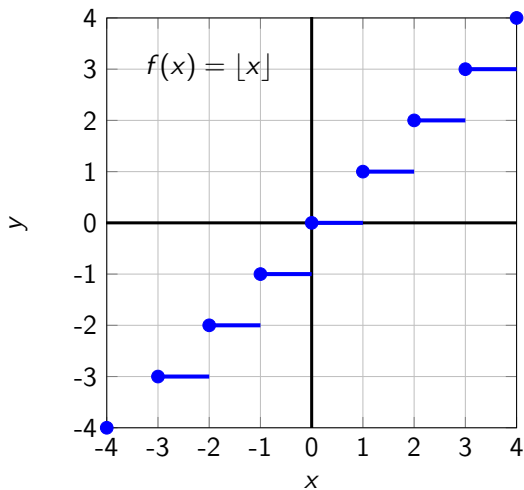
- ▶ The integer part function  $f(x) = \lfloor x \rfloor$  is *monotone increasing*, like the upper integer part function  $f(x) = \lceil x \rceil$ .

## Graph of $f(x) = \lceil x \rceil$

Graphs of the functions  $f(x) = \lceil x \rceil$  and  $f(x) = \lfloor x \rfloor$  on the interval  $[-4, 4]$ .

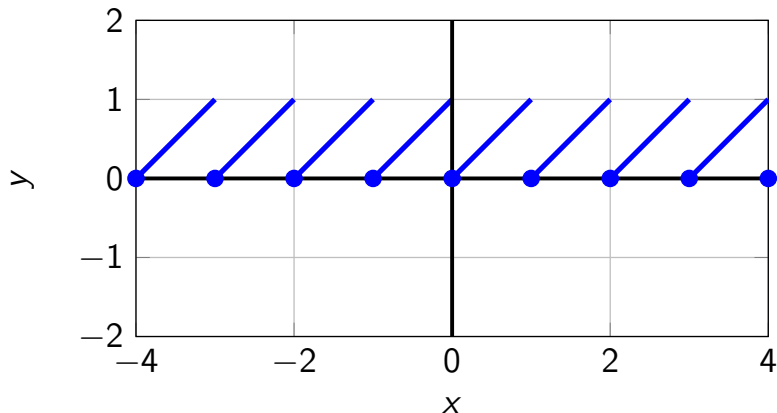


Graph of  $f(x) = \lfloor x \rfloor$



# Graph of the fractional part

Note that the fractional part is a periodic function with period 1.



# Summary

## Integer part and fractional part:

- ▶ Integer part  $\lfloor x \rfloor$  and upper integer part  $\lceil x \rceil$ : functions from  $\mathbb{R}$  to  $\mathbb{Z}$ , monotone increasing, piecewise constant (“step” graphs)
- ▶ Fractional part  $\{x\}$ : function from  $\mathbb{R}$  to  $[0, 1)$ , periodic with period 1
- ▶ They are typical functions in the writing of algorithms

# Piecewise-defined functions



# Introduction: Piecewise-defined functions

- ▶ Starting from the elementary functions, new ones can be built by using **different analytic definitions on different intervals**
- ▶ In this section we give the definition of piecewise-defined function and we see an example with its graph
- ▶ Piecewise functions are widely used to model situations in which the “rule” changes depending on the values of the variable

# Piecewise-defined functions

## Definition 4: of piecewise-defined function

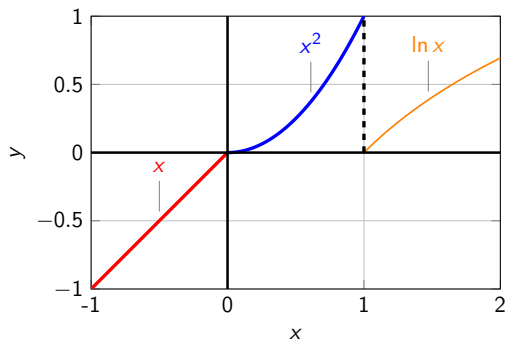
*A function  $f$  whose value  $f(x)$  is computed through different “instructions” depending on the interval in which  $x$  lies is called a **piecewise-defined function**.*

## Example

Consider for example

$$f(x) = \begin{cases} \ln x & \text{if } x > 1, \\ x^2 & \text{if } 0 < x \leq 1, \\ x & \text{if } x \leq 0 \end{cases}$$

its graph is:



# Piecewise functions and algorithms

- ▶ From the *logical / algorithmic* point of view, one can say that a function of this kind has the peculiarity of being built using not only mathematical functions but also the **logical function** “if ... then”.



## Piecewise-defined functions:

- ▶ The value  $f(x)$  is computed with different “instructions” depending on the interval in which  $x$  lies
- ▶ They are built starting from the elementary functions
- ▶ Besides mathematical functions, they use the logical function “if . . . then”

# Summary

- ▶ **Integer part**  $\lfloor x \rfloor$  and **upper integer part**  $\lceil x \rceil$ : step functions, monotone increasing, with values in  $\mathbb{Z}$
- ▶ **Fractional part**  $\{x\}$ : values in  $[0, 1)$ , periodic with period 1
- ▶ **Piecewise-defined functions**: different analytic definitions on different intervals, built with the “if ... then” logic